Chapter 13

Plant Growth and Development

Where and how do roots, stems, leaves, flowers, fruits and seeds arise, and in an orderly sequence? Development is the sum of two processes, growth and differentiation.

The questions this chapter opens with

- trees increase in height or girth over time, while the leaves, flowers and fruits of the same tree have limited dimensions and appear and fall periodically
- all plant organs are made up of a variety of tissues, and there is a relationship between the structure of a cell, a tissue, an organ and the function they perform
- all cells of a plant are descendents of the zygote, so why and how do they come to have different structural and functional attributes?

An ordered succession

- a mature plant develops from the zygote by a precise and highly ordered succession of events
- a complex body organisation is formed that produces roots, leaves, branches, flowers, fruits and seeds, and eventually they die
- the first step in the process of plant growth is seed germination, which happens when favourable conditions for growth exist in the environment
- without them the seeds do not germinate and go into suspended growth or rest, resuming metabolic activities once favourable conditions return

Intrinsic and extrinsic

- this chapter also covers some of the factors which govern and control these developmental processes
- these factors are both intrinsic, that is internal, and extrinsic, that is external, to the plant

In this chapter

13.1 Growth

13.5 Growth rates

13.9 Differentiation

13.10 Development

13.12 Plant growth regulators

13.14 Auxins to abscisic acid

Seed coat splitting as the seedling emerges

Hypocotyl arching to form the epicotyl hook

Cotyledons carried above the soil line

Epicotyl extending above them

Figure 13.A Germination and seedling development in bean

13.1 Growth

What growth is

- growth is regarded as one of the most fundamental and conspicuous characteristics of a living being
- it can be defined as an irreversible permanent increase in size of an organ or its parts, or even of an individual cell
- it is generally accompanied by metabolic processes, both anabolic and catabolic, that occur at the expense of energy, so that the expansion of a leaf is growth

13.2 Growth is generally indeterminate

The open form of growth

- plant growth is unique because plants retain the capacity for unlimited growth throughout their life, due to the presence of meristems at certain locations in their body
- the cells of such meristems have the capacity to divide and self-perpetuate, while the product soon loses the capacity to divide and makes up the plant body
- this form of growth, wherein new cells are always being added to the plant body by the activity of the meristem, is called the open form of growth

Shoot apical meristem at the top of the shoot

Root apical meristem at the tip of the root

Vascular cambium in shoot and in root

Arrows showing the direction of growth

Figure 13.B Diagrammatic representation of locations of root apical meristem, shoot aplical meristem and vascular cambium. Arrows exhibit the direction of growth of cells and organ

Parallel lines marked behind the root apex

Zones A, B, C and D lying immediately behind it

Those zones show the greatest separation after growth

Figure 13.C Detection of zones of elongation by the parallel line technique. Zones A, B, C, D immediately behind the apex have elongated most.

Primary and secondary growth

- the root apical meristem and the shoot apical meristem are responsible for the primary growth of plants, and principally contribute to elongation along their axis
- in dicotyledonous plants and gymnosperms the lateral meristems, vascular cambium and cork-cambium, appear later in life
- these are the meristems that cause the increase in the girth of the organs in which they are active, which is known as secondary growth

13.3 Growth is measurable

What is measured

- at a cellular level growth is principally a consequence of increase in the amount of protoplasm, which is difficult to measure directly, so one measures a quantity more or less proportional to it
- growth is measured by increase in fresh weight, dry weight, length, area, volume and cell number
- one single maize root apical meristem can give rise to more than 17,500 new cells per hour, whereas cells in a watermelon may increase in size by up to 3,50,000 times
- in the former growth is expressed as increase in cell number, in the latter as increase in size of the cell; the growth of a pollen tube is measured by its length, while an increase in surface area denotes growth in a dorsiventral leaf

13.4 Phases of growth

The meristematic phase

- the period of growth is generally divided into three phases, meristematic, elongation and maturation
- the constantly dividing cells, both at the root apex and the shoot apex, represent the meristematic phase
- the cells in this region are rich in protoplasm and possess large conspicuous nuclei
- their cell walls are primary in nature, thin and cellulosic, with abundant plasmodesmatal connections

Arithmetic growth, a straight line against time

Geometric growth, a curve rising ever more steeply

Stages during embryo development, showing both phases in turn

Figure 13.D Diagrammatic representation of : (a) Arithmetic (b) Geometric growth and (c) Stages during embryo development showing geometric and arithematic phases

Elongation and maturation

- the cells proximal, that is just next and away from the tip, to the meristematic zone represent the phase of elongation
- increased vacuolation, cell enlargement and new cell wall deposition are the characteristics of cells in this phase
- further away from the apex, more proximal to the phase of elongation, lies the portion of the axis undergoing the phase of maturation
- the cells of this zone attain their maximal size in terms of wall thickening and protoplasmic modifications, and most of the tissues and cell types studied earlier represent this phase

13.5 Growth rates

Rate and its forms

- the increased growth per unit time is termed the growth rate, and can be expressed mathematically
- an organism, or a part of an organism, can produce more cells in a variety of ways
- the growth rate shows an increase that may be arithmetic or geometrical

Length plotted against time

A straight line, since the rate is constant

Figure 13.E Constant linear growth, a plot of length L against time t

Lag phase, where growth is slow at first

Exponential phase, where it rises rapidly

Stationary phase, where limited nutrients slow it again

Figure 13.F An idealised sigmoid growth curve typical of cells in culture, and many higher plants and plant organs

13.6 Two kinds of growth

Arithmetic growth

- following mitotic cell division only one daughter cell continues to divide, while the other differentiates and matures
- the simplest expression is a root elongating at a constant rate, and plotting the length of the organ against time gives a linear curve
- it is expressed as Lt = L0 + rt, where Lt is length at time t, L0 is length at time zero, and r is the growth rate or elongation per unit time

Geometric growth

- in most systems the initial growth is slow, the lag phase, and increases rapidly thereafter at an exponential rate, the log or exponential phase
- here both progeny cells following mitotic division retain the ability to divide and continue to do so
- with limited nutrient supply the growth slows down, leading to a stationary phase, and plotting the parameter of growth against time gives a typical sigmoid or S-curve
- a sigmoid curve is a characteristic of living organisms growing in a natural environment, and is typical for all cells, tissues and organs of a plant

The exponential expression

- exponential growth is expressed as W1 = W0 multiplied by e raised to the power rt
- W1 is the final size in weight, height or number, W0 the initial size at the beginning of the period, r the growth rate, t the time of growth, and e the base of natural logarithms
- here r is the relative growth rate, and is also the measure of the ability of the plant to produce new plant material, referred to as efficiency index
- hence the final size W1 depends on the initial size W0

Leaf A and leaf B, of different starting sizes

Both increase their area by the same amount in the same time

Giving leaves A1 and B1

The smaller leaf shows the higher relative growth rate

Figure 13.G Diagrammatic comparison of absolute and relative growth rates. Both leaves A and B have increased their area by 5 cm2 in a given time to produce A1, B1 leaves.

13.7 Absolute and relative growth rates

Two ways to compare

- measurement and the comparison of total growth per unit time is called the absolute growth rate
- the growth of the given system per unit time expressed on a common basis, for example per unit initial parameter, is called the relative growth rate

13.8 Conditions for growth

Water, oxygen and nutrients

- plant cells grow in size by cell enlargement, which in turn requires water, and turgidity of cells helps in extension growth, so plant growth and further development is intimately linked to the water status of the plant
- water also provides the medium for enzymatic activities needed for growth
- oxygen helps in releasing metabolic energy essential for growth activities
- nutrients, both macro and micro essential elements, are required for the synthesis of protoplasm and act as a source of energy

Temperature and signals

- every plant organism has an optimum temperature range best suited for its growth
- any deviation from this range could be detrimental to its survival
- environmental signals such as light and gravity also affect certain phases and stages of growth

13.9 Differentiation, dedifferentiation and redifferentiation

Differentiation

- the cells derived from root apical and shoot apical meristems and cambium differentiate and mature to perform specific functions, and this act leading to maturation is termed differentiation
- during differentiation, cells undergo few to major structural changes both in their cell walls and protoplasm
- to form a tracheary element, for example, the cells lose their protoplasm and develop very strong, elastic, lignocellulosic secondary cell walls, to carry water to long distances even under extreme tension

Dedifferentiation and redifferentiation

- living differentiated cells that have lost the capacity to divide can regain that capacity under certain conditions, a phenomenon termed dedifferentiation
- an example is the formation of the meristems interfascicular cambium and cork cambium from fully differentiated parenchyma cells
- such meristems are able to divide and produce cells that once again lose the capacity to divide but mature to perform specific functions, that is they get redifferentiated

Differentiation is open too

- growth in plants is open, that is it can be indeterminate or determinate, and differentiation in plants is open as well, because cells and tissues arising out of the same meristem have different structures at maturity
- the final structure at maturity of a cell or tissue is also determined by the location of the cell within the organ
- cells positioned away from root apical meristems differentiate as root-cap cells, while those pushed to the periphery mature as epidermis

13.10 Development

The whole life cycle

- development is a term that includes all changes an organism goes through during its life cycle, from germination of the seed to senescence
- the sequence of processes which constitute the development of a cell of a higher plant is also applicable to tissues and organs

Meristematic cell dividing

Plasmatic growth, then expansion or elongation

Differentiation and maturation to a mature cell

Senescence, and finally death

Figure 13.H Sequence of the developmental process in a plant cell

Larkspur, whose juvenile leaves differ in shape from mature ones

Buttercup, whose leaves differ in air and in water

Figure 13.I Heterophylly in (a) larkspur and (b) buttercup

13.11 Plasticity

Different pathways, different structures

- plants follow different pathways in response to environment or phases of life to form different kinds of structures, and this ability is called plasticity
- heterophylly in cotton, coriander and larkspur is an example, where the leaves of the juvenile plant differ in shape from those of the mature plant
- the difference in shapes of leaves produced in air and those produced in water in buttercup also represents heterophyllous development due to environment
- this phenomenon of heterophylly is an example of plasticity

A coleoptile with its tip intact, bending towards the light

A coleoptile with the tip removed, which does not bend

A coleoptile with the tip covered, which does not bend

Arrows indicating the direction of light

Figure 13.K Experiment used to demonstrate that tip of the coleoptile is the source of auxin. Arrows indicate direction of light

How the three relate

- growth, differentiation and development are very closely related events in the life of a plant, development being considered the sum of growth and differentiation
- development in plants is under the control of intrinsic and extrinsic factors
- the former includes both intracellular genetic factors and intercellular factors such as plant growth regulators, while the latter includes light, temperature, water, oxygen and nutrition

13.12 Plant growth regulators

What they are

- plant growth regulators are small, simple molecules of diverse chemical composition
- they could be indole compounds such as indole-3-acetic acid, adenine derivatives such as kinetin, derivatives of carotenoids such as abscisic acid, terpenes such as gibberellic acid, or gases such as ethylene
- they are variously described as plant growth substances, plant hormones or phytohormones

Two groups

- one group is involved in growth promoting activities such as cell division, cell enlargement, pattern formation, tropic growth, flowering, fruiting and seed formation, and these are called plant growth promoters, namely auxins, gibberellins and cytokinins
- the other group plays an important role in plant responses to wounds and stresses of biotic and abiotic origin, and in growth inhibiting activities such as dormancy and abscission, to which abscisic acid belongs
- the gaseous regulator ethylene could fit either group, but it is largely an inhibitor of growth activities

13.13 The discovery of the regulators

Auxin

- the discovery of each of the five major groups has been accidental
- it started with Charles Darwin and his son Francis observing that the coleoptiles of canary grass responded to unilateral illumination by growing towards the light source, that is phototropism
- after a series of experiments it was concluded that the tip of the coleoptile was the site of a transmittable influence that caused the bending of the entire coleoptile, and auxin was isolated by F.W. Went from tips of coleoptiles of oat seedlings

Gibberellins and cytokinins

- the bakanae or foolish seedling disease of rice seedlings was caused by the fungal pathogen Gibberella fujikuroi, and E. Kurosawa (1926) reported the appearance of the symptoms in rice seedlings treated with sterile filtrates of the fungus, the active substances later identified as gibberellic acid
- F. Skoog and co-workers observed that from internodal segments of tobacco stems the callus, a mass of undifferentiated cells, proliferated only if, in addition to auxins, the nutrient medium was supplemented with extracts of vascular tissues, yeast extract, coconut milk or DNA
- Miller and co-workers (1955) later identified and crystallised the cytokinesis promoting active substance, which they termed kinetin

Abscisic acid and ethylene

- during the mid-1960s three independent researches reported the purification and chemical characterisation of three different kinds of inhibitors, inhibitor-B, abscission II and dormin, all three later proved to be chemically identical and named abscisic acid
- H.H. Cousins (1910) confirmed the release of a volatile substance from ripened oranges that hastened the ripening of stored unripened bananas
- this volatile substance was later identified as ethylene, a gaseous regulator

13.14 Auxins

What they are and where from

- auxins, from the Greek auxein meaning to grow, were first isolated from human urine
- the term is applied to indole-3-acetic acid and to other natural and synthetic compounds having certain growth regulating properties
- they are generally produced by the growing apices of the stems and roots, from where they migrate to the regions of their action
- auxins such as IAA and indole butyric acid have been isolated from plants, while naphthalene acetic acid and 2,4-D are synthetic auxins, all used extensively in agricultural and horticultural practices

What they do

- they help to initiate rooting in stem cuttings, an application widely used for plant propagation, and promote flowering as in pineapples
- they help to prevent fruit and leaf drop at early stages, but promote the abscission of older mature leaves and fruits
- in most higher plants the growing apical bud inhibits the growth of the lateral or axillary buds, a phenomenon called apical dominance
- removal of shoot tips, that is decapitation, usually results in the growth of lateral buds, and this is widely applied in tea plantations and hedge-making

A plant with the apical bud intact, lateral buds suppressed

A plant with the apical bud removed

Lateral buds growing into branches after decapitation

Figure 13.L Apical dominance in plants : (a) A plant with apical bud intact (b) A plant with apical bud removed Note the growth of lateral buds into branches after decapitation.

More uses of auxins

- auxins also induce parthenocarpy, as in tomatoes, and are widely used as herbicides
- 2,4-D, widely used to kill dicotyledonous weeds, does not affect mature monocotyledonous plants, and is used to prepare weed-free lawns by gardeners
- auxin also controls xylem differentiation and helps in cell division

13.15 Gibberellins

More than a hundred of them

- there are more than 100 gibberellins reported from widely different organisms such as fungi and higher plants, denoted GA1, GA2, GA3 and so on
- gibberellic acid was one of the first to be discovered and remains the most intensively studied form, and all of them are acidic
- their ability to cause an increase in length of axis is used to increase the length of grape stalks, and they cause fruits such as apple to elongate and improve their shape
- they also delay senescence, so fruits can be left on the tree longer to extend the market period, and gibberellic acid is used to speed up the malting process in the brewing industry

In the field

- sugarcane stores carbohydrate as sugar in its stems, and spraying the crop with gibberellins increases the length of the stem, increasing the yield by as much as 20 tonnes per acre
- spraying juvenile conifers with gibberellins hastens the maturity period, leading to early seed production
- they also promote bolting, that is internode elongation just prior to flowering, in beet, cabbages and many plants with rosette habit

13.16 Cytokinins

From kinetin to zeatin

- cytokinins have specific effects on cytokinesis, and were discovered as kinetin, a modified form of adenine, from autoclaved herring sperm DNA
- kinetin does not occur naturally in plants, and the search for natural substances with cytokinin-like activity led to the isolation of zeatin from corn kernels and coconut milk
- since then several naturally occurring cytokinins, and some synthetic compounds with cell division promoting activity, have been identified

What cytokinins do

- natural cytokinins are synthesised in regions where rapid cell division occurs, for example root apices, developing shoot buds and young fruits
- they help to produce new leaves, chloroplasts in leaves, lateral shoot growth and adventitious shoot formation
- they help overcome apical dominance, and promote nutrient mobilisation, which helps in the delay of leaf senescence

13.17 Ethylene

A simple gas

- ethylene is synthesised in large amounts by tissues undergoing senescence and by ripening fruits
- its influences include horizontal growth of seedlings, swelling of the axis and apical hook formation in dicot seedlings
- it promotes senescence and abscission of plant organs, especially of leaves and flowers, and is highly effective in fruit ripening, enhancing the respiration rate during ripening, a rise called respiratory climactic
- it breaks seed and bud dormancy, initiates germination in peanut seeds and sprouting of potato tubers

In agriculture

- it promotes rapid internode and petiole elongation in deep water rice plants, helping leaves and upper parts of the shoot to remain above water
- it also promotes root growth and root hair formation, helping plants increase their absorption surface
- it is used to initiate flowering and for synchronising fruit-set in pineapples, and induces flowering in mango
- the most widely used source is ethephon, which in aqueous solution is readily absorbed and transported within the plant and releases ethylene slowly, hastening fruit ripening in tomatoes and apples, accelerating abscission in flowers and fruits, and promoting female flowers in cucumbers

13.18 Abscisic acid

The stress hormone

- abscisic acid was discovered for its role in regulating abscission and dormancy, but like other regulators it has other wide ranging effects on plant growth and development
- it acts as a general plant growth inhibitor and an inhibitor of plant metabolism, and inhibits seed germination
- it stimulates the closure of stomata and increases the tolerance of plants to various kinds of stresses, which is why it is also called the stress hormone

Abscisic acid, concluded

- it plays an important role in seed development, maturation and dormancy
- by inducing dormancy it helps seeds withstand desiccation and other factors unfavourable for growth
- in most situations it acts as an antagonist to the gibberellins

How the regulators interact

- for any and every phase of growth, differentiation and development, one or the other regulator has some role to play, and such roles could be complimentary or antagonistic, individualistic or synergistic
- there are a number of events in the life of a plant where more than one regulator interacts to affect that event, such as dormancy in seeds and buds, abscission, senescence and apical dominance
- the role of the regulators is only one kind of intrinsic control, and along with genomic control and extrinsic factors they play an important role in plant growth and development
- many extrinsic factors such as temperature and light control plant growth and development via the regulators, in events such as vernalisation, flowering, dormancy, seed germination and plant movements

13.19 Summary

Growth

- growth is an irreversible increase expressed in parameters such as size, area, length, height, volume and cell number, and conspicuously involves increased protoplasmic material
- in plants, meristems are the sites of growth, root and shoot apical meristems, sometimes along with intercalary meristem, contributing to the elongation growth of plant axes, and growth is indeterminate in higher plants
- following cell division in root and shoot apical meristem cells the growth could be arithmetic or geometrical, and is generally not sustained at a high rate throughout the life of the cell, tissue, organ or organism
- one can define three principal phases of growth, the lag, the log and the senescent phase

Differentiation and development

- when a cell loses the capacity to divide, it leads to differentiation, which results in development of structures commensurate with the function the cell finally has to perform
- general principles for differentiation of cells, tissues and organs are similar, and a differentiated cell may dedifferentiate and then redifferentiate
- since differentiation in plants is open, the development could also be flexible, development being the sum of growth and differentiation, and plants exhibit plasticity in development

The regulators

- plant growth and development are under the control of both intrinsic and extrinsic factors, the intercellular intrinsic factors being the chemical substances called plant growth regulators
- there are diverse groups of them, principally belonging to five: auxins, gibberellins, cytokinins, abscisic acid and ethylene
- they are synthesised in various parts of the plant and control different differentiation and developmental events, any one having diverse physiological effects while diverse regulators also manifest similar effects
- they may act synergistically or antagonistically, and plant growth and development is also affected by light, temperature, nutrition, oxygen status, gravity and other external factors

13.20 Exercises

From this chapter

1. Define growth, differentiation, development, dedifferentiation, redifferentiation, determinate growth, meristem and growth rate.
2. Why is no single parameter good enough to demonstrate growth throughout the life of a flowering plant?
3. Describe briefly: (a) arithmetic growth (b) geometric growth (c) the sigmoid growth curve (d) absolute and relative growth rates
4. List five main groups of natural plant growth regulators. Write a note on the discovery, physiological functions and agricultural or horticultural applications of any one of them.
5. Why is abscisic acid also known as the stress hormone?
6. Comment on the statement that both growth and differentiation in higher plants are open.
7. Explain how both a short day plant and a long day plant can flower simultaneously in a given place.

From this chapter, continued

1. Which plant growth regulator would you use if you were asked to: (a) induce rooting in a twig (b) quickly ripen a fruit (c) delay leaf senescence (d) induce growth in axillary buds (e) bolt a rosette plant (f) induce immediate stomatal closure in leaves
2. Would a defoliated plant respond to a photoperiodic cycle? Why?
3. What would be expected to happen if: (a) gibberellic acid is applied to rice seedlings (b) dividing cells stop differentiating (c) a rotten fruit gets mixed with unripe fruits (d) you forget to add cytokinin to the culture medium